

## COMPSCI 466: Homework 6

**Due date:** May 5 11:59pm.

**Problem 1. (100 points.)** Define the symmetric encryption scheme  $\text{SE} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$  where  $\mathcal{K}$  returns a uniformly random 128-bit key  $K$ , and the encryption and decryption algorithms are as follows.

**Algorithm  $\mathcal{E}_K(M)$ :**

```

If  $|M| \neq 512$  then Ret  $\perp$ 
Parse  $M$  as  $M[1]||M[2]||M[3]||M[4]$ 
 $C_e[0] \leftarrow \{0, 1\}^{128}$ 
 $C_m[0] \leftarrow 0^{128}$ 
For  $i = 1, 2, 3, 4$  do:
     $C_e[i] \leftarrow \text{AES}_K(C_e[i-1] \oplus M[i])$ 
     $C_m[i] \leftarrow \text{AES}_K(C_m[i-1] \oplus M[i])$ 
 $C_e \leftarrow C_e[0]||C_e[1]||C_e[2]||C_e[3]||C_e[4]$ 
 $T \leftarrow C_m[4]$ 
Ret  $(C_e, T)$ 

```

**Algorithm  $\mathcal{D}_K((C_e, T))$ :**

```

If  $|C_e| \neq 640$  then Ret  $\perp$ 
Parse  $C_e$  as  $C_e[0]||C_e[1]||\dots||C_e[4]$ 
 $C_m[0] \leftarrow 0^{128}$ 
For  $i = 1, 2, 3, 4$  do:
     $M[i] \leftarrow \text{AES}_K^{-1}(C_e[i]) \oplus C_e[i-1]$ 
     $C_m[i] \leftarrow \text{AES}_K(C_m[i-1] \oplus M[i])$ 
If  $C_m[4] = T$  then Ret  $M[1]||M[2]||M[3]||M[4]$ 
Else Ret  $\perp$ 

```

**(Part A — 40 points.)** Show that  $\text{SE}$  is *not* IND-CPA secure. State your adversary  $A$  in pseudocode and analyze its advantage and resource usage.

**(Part B — 40 points.)** Show that  $\text{SE}$  is *not* INT-CTXT secure. State your adversary  $A$  in pseudocode and analyze its advantage and resource usage.

**(Part C — 20 points.)** Is  $\text{SE}$  an Encrypt-and-MAC scheme? Briefly justify your answer.

**Bonus Problem. (40 points.)** Consider the modified scheme  $\text{SE}' = (\mathcal{K}', \mathcal{E}', \mathcal{D}')$  where  $\mathcal{K}'$  samples  $K_e, K_m \leftarrow \{0, 1\}^{128}$  and returns  $(K_e, K_m)$ . The encryption and decryption algorithms are:

**Algorithm  $\mathcal{E}'_{(K_e, K_m)}(M)$ :**

```

If  $|M| \neq 512$  then Ret  $\perp$ 
Parse  $M$  as  $M[1]||M[2]||M[3]||M[4]$ 
 $C_e[0] \leftarrow \{0, 1\}^{128}$ 
 $C_m[0] \leftarrow 0^{128}$ 
For  $i = 1, 2, 3, 4$  do:
     $C_e[i] \leftarrow \text{AES}_{K_e}(C_e[i-1] \oplus M[i])$ 
     $C_m[i] \leftarrow \text{AES}_{K_m}(C_m[i-1] \oplus M[i])$ 
 $C_e \leftarrow C_e[0]||C_e[1]||C_e[2]||C_e[3]||C_e[4]$ 
 $T \leftarrow C_m[4]$ 
Ret  $(C_e, T)$ 

```

**Algorithm  $\mathcal{D}'_{(K_e, K_m)}((C_e, T))$ :**

```

If  $|C_e| \neq 640$  then Ret  $\perp$ 
Parse  $C_e$  as  $C_e[0]||C_e[1]||\dots||C_e[4]$ 
 $C_m[0] \leftarrow 0^{128}$ 
For  $i = 1, 2, 3, 4$  do:
     $M[i] \leftarrow \text{AES}_{K_e}^{-1}(C_e[i]) \oplus C_e[i-1]$ 
     $C_m[i] \leftarrow \text{AES}_{K_m}(C_m[i-1] \oplus M[i])$ 
If  $C_m[4] = T$  then Ret  $M[1]||M[2]||M[3]||M[4]$ 
Else Ret  $\perp$ 

```

**(Part A — 20 points.)** Is  $SE'$  IND-CPA secure? If yes, give convincing intuition. If no, give an attack.

**(Part B — 20 points.)** Is  $SE'$  INT-CTXT secure? If yes, give convincing intuition. If no, give an attack.